

CRYPTOCURRENCY MARKET DYNAMICS

CS4379G

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01

Motivation

Motivation

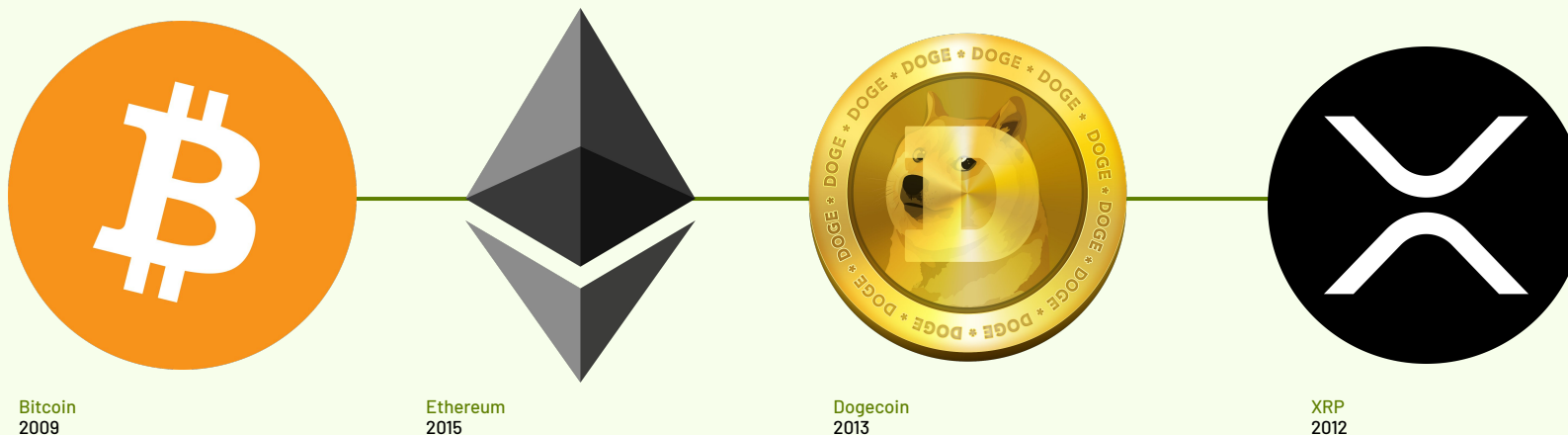


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- Interest in cryptocurrency and investment culture
- Curiosity about volatility and rapid market changes
- Compare behavior across major cryptocurrencies
- Use interactive visualizations to explore market patterns

02

Data Description



Data Description

- The dataset used for this project was obtained from Kaggle and included historical market data for Bitcoin, Ethereum, Dogecoin, and XRP.
- The data contained information such as opening and closing prices, highs, lows, trading volume, and dates.
- These cryptocurrencies were selected because they are among the more well-known and influential cryptocurrencies at different points in time, while also displaying different levels of volatility and market behavior throughout the analysis.

03

Methodology

Methodology

General Workflow

Data Collection

- Imported CSV datasets via Kaggle using `pd.read_csv()`
- Bitcoin, Ethereum, XRP, Dogecoin
- Columns used Date, Open, Close, High, Low, Volume

Analysis

- Calculated daily returns using `pct_change()`
- Created price, volume, heatmap and boxplot visualizations
- Compared volatility and market behaviors

Data Cleaning

- Organized datasets chronologically
- Converted dates using `pd.to_datetime()`
- Prepped close pricing for return and volatility calculations

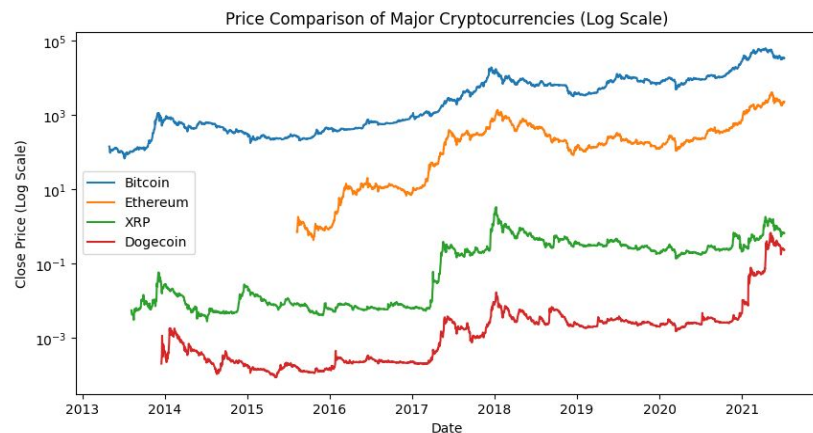
Dashboard

- Created Streamlit dashboard
- Implemented cryptocurrency selection using `st.selectbox()`
- Displayed Price Over Time and Daily Returns

04

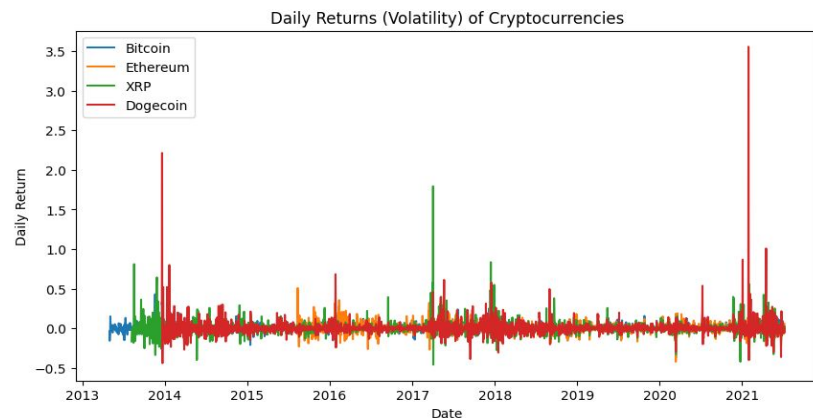
Key Findings

Key Findings



Price Comparison (Log)

- Bitcoin's large price overwhelmed the others, which made log scaling needed in order to have clear visuals
- All the cryptocurrencies roughly followed similar growth and recession patterns despite the large difference in price scales
- 2017-2018 showed the largest spikes and general growth



Daily Returns (Volatility)

- Bitcoin remained the most stable overall despite its larger market
- XRP and Dogecoin were by far the most volatile, with Dogecoin almost doubling the second largest spike
- Ethereum was its most volatile at the start and then remained fairly stable afterwards in comparison

05

Limitations

Limitations

- Limited cryptocurrencies were analyzed, which then limited representation of the vast crypto market
- Dataset is not updated to current day, when trending ideas may be at potentially its most influential time
- Focused on historical trends rather than predicting future market behavior

